9.8.19

AI25BTECH11017-BALU

Question:

1) Two circles $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$ are given. Then the equation of the circle through their points of intersection and the point $(1, 1)$ is (1980)
   (a) $x^2 + y^2 - 6x + 4 = 0$
   (b) $x^2 + y^2 - 3x + 1 = 0$
   (c) $x^2 + y^2 - 4y + 2 = 0$
   (d) none of these

Solution:
Let us solve the given equation theoretically and then verify the solution computationally
According to the question,
equation circles are

$$\mathbf{X}^T\mathbf{X} - 6 = 0 \tag{1.1}$$

$$\mathbf{X}^T\mathbf{X} + 2\begin{pmatrix} -3 & 0 \end{pmatrix}\mathbf{X} + 8 = 0 \tag{1.2}$$

substute 1.1 in 1.2

$$\begin{pmatrix} -3 & 0 \end{pmatrix}\mathbf{X} = -7 \tag{1.3}$$

$$x = \frac{7}{3} \tag{1.4}$$

substute 1.4 in 1.1

$$y = \frac{\sqrt{5}}{3}, \frac{-\sqrt{5}}{3} \tag{1.5}$$

let equation of requried circle is

$$\mathbf{X}^T\mathbf{X} + 2\mathbf{u}^T\mathbf{X} + f = 0 \tag{1.6}$$

$$\tag{1.7}$$

$$\begin{pmatrix} 2 & 14 & 14 \\ 2 & \frac{2\sqrt{5}}{3} & \frac{-2\sqrt{5}}{3} \\ 1 & 1 & 1 \end{pmatrix}^T \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} -2 \\ -6 \\ -6 \end{pmatrix} \tag{1.8}$$

$$f = 1 \tag{1.9}$$

$$\mathbf{u} = \begin{pmatrix} \frac{-3}{2} \\ 0 \end{pmatrix} \tag{1.10}$$

the reqiured circle equation is

$$x^2 + y^2 - 3x + 1 = 0 \tag{1.11}$$

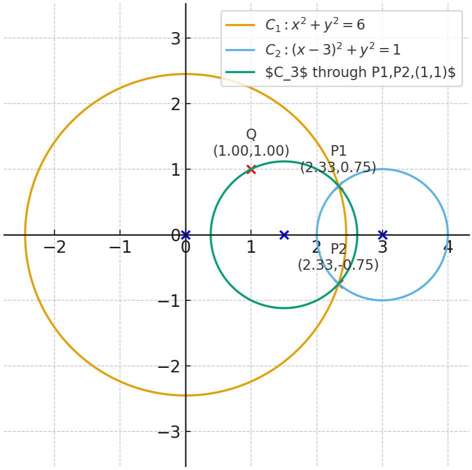

Fig. 1.1